



Independent QCE practice

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Practice Set A

2025 SYLLABUS EDITION

Multiple choice question book

Chemistry

Paper 1

General instruction

- Work in this book will not be marked.

Section 1

Instruction

- Respond to these questions in the Paper 1 question and response book.
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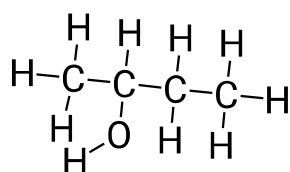
QUESTION 1

A sealed vessel contains a reversible reaction at dynamic equilibrium. Which statement describes the reaction?

- (A) Both reactions have stopped and the reactant and product concentrations are equal.
- (B) The forward and reverse reactions continue at equal, non-zero rates.
- (C) The forward reaction continues while the reverse reaction has stopped.
- (D) Reactant and product particles no longer collide.

QUESTION 2

Which statement correctly identifies the functional group and compound class represented by the displayed molecule?



- (A) Its functional-group name is alcohol and its class name is hydroxyl.
- (B) Its functional-group name is carbonyl and its class name is aldehyde.
- (C) Its functional-group name is carboxyl and its class name is carboxylic acid.
- (D) Its functional-group name is hydroxyl and its class name is alcohol.

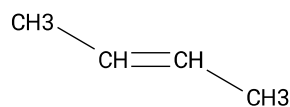
QUESTION 3

Calcium carbonate decomposes according to $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$. Determine the equilibrium-law expression.

- (A) $K_c = [\text{CO}_2]$
- (B) $K_c = \frac{[\text{CaO}][\text{CO}_2]}{[\text{CaCO}_3]}$
- (C) $K_c = [\text{CaO}][\text{CO}_2]$
- (D) $K_c = \frac{1}{[\text{CO}_2]}$

QUESTION 4

Determine the name of the displayed geometrical isomer.



- (A) cis-but-1-ene
- (B) trans-but-1-ene
- (C) trans-but-2-ene
- (D) cis-but-2-ene

QUESTION 5

Two monoprotic acids have the same analytical concentration and temperature. Use the data to determine the supported conclusion.

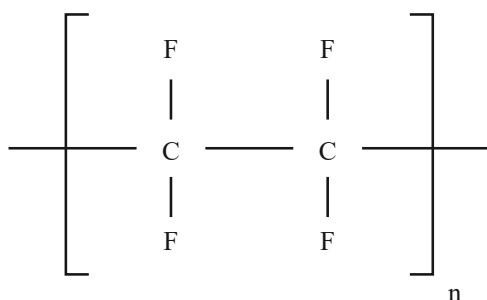
Acid	pH	Conductivity
P	2.00	high
Q	3.18	low

Acid evidence

- (A) Acid Q is stronger because it has the higher pH.
- (B) Acid P is stronger because it has the lower pH and higher conductivity.
- (C) Acid P is more concentrated because it has the lower pH.
- (D) Acid Q is stronger because it has the lower conductivity.

QUESTION 6

The repeating unit of an addition polymer is displayed. Determine the monomer used to make the polymer.



- (A) tetrafluoroethene
- (B) 1,1-difluoroethane
- (C) ethene
- (D) tetrafluoroethane

QUESTION 7

For $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, $K_c = 50.0$. A mixture has $[\text{H}_2] = 0.200 \text{ M}$, $[\text{I}_2] = 0.200 \text{ M}$ and $[\text{HI}] = 0.500 \text{ M}$. Predict the net reaction.

- (A) The system is at equilibrium.
- (B) The reverse reaction is favoured because $Q > K_c$.
- (C) The reverse reaction is favoured because $Q < K_c$.
- (D) The forward reaction is favoured because $Q < K_c$.

QUESTION 8

2.00 mol of N_2 reacts with 4.50 mol of H_2 according to $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$. The actual mass of ammonia collected is 38.3 g. Determine the percentage yield.

- (A) 50.0%
- (B) 56.2%
- (C) 74.9%
- (D) 133%

QUESTION 9

An indicator obeys $\text{HIn}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{In}^-(\text{aq})$. Which condition applies when $\text{pH} = \text{p}K_a$ for the indicator?

- (A) $[\text{HIn}] = [\text{In}^-]$
- (B) $[\text{H}^+] = [\text{In}^-]$
- (C) $[\text{HIn}] = 0$
- (D) $\text{pH} = 7$

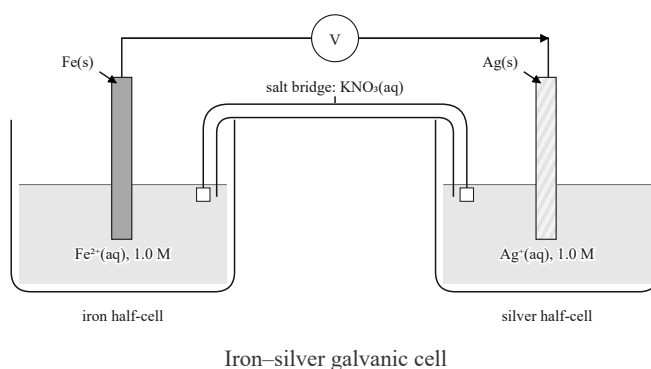
QUESTION 10

Determine the major product when pent-1-ene reacts with hydrogen bromide.

- (A) 1-bromopentane
- (B) 2-bromopentane
- (C) 3-bromopentane
- (D) 2-bromo-2-methylbutane

QUESTION 11

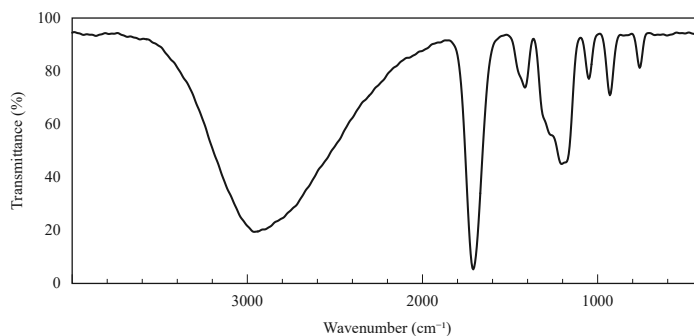
The standard reduction potentials are $E^\circ(\text{Ag}^+/\text{Ag}) = +0.80 \text{ V}$ and $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V}$. Determine the standard potential of the displayed galvanic cell.



- (A) -1.24 V
- (B) $+0.36 \text{ V}$
- (C) $+1.24 \text{ V}$
- (D) $+1.60 \text{ V}$

QUESTION 12

Use the infrared spectrum to determine the class of the unknown organic compound.

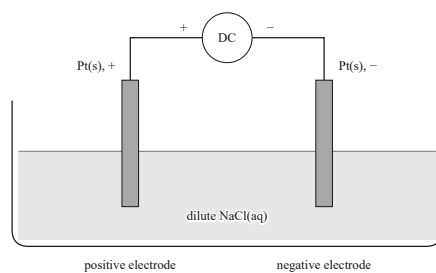


Infrared spectrum of the unknown

- (A) alkene
- (B) alcohol
- (C) amine
- (D) carboxylic acid

QUESTION 13

Dilute aqueous sodium chloride is electrolysed using inert electrodes. Determine the main product at the positive electrode.



Electrolysis of dilute sodium chloride

- (A) $\text{H}_2(\text{g})$
- (B) $\text{O}_2(\text{g})$
- (C) $\text{Na}(\text{l})$
- (D) $\text{Cl}_2(\text{g})$

QUESTION 14

Use the boiling-point data to determine the best explanation for the trend.

Alkane	Boiling point (°C)
pentane	36
hexane	69
heptane	98

Boiling points of three alkanes

- (A) Longer alkane chains have larger electron clouds and stronger dispersion forces.
- (B) Longer alkane chains form hydrogen bonds with one another.
- (C) Shorter alkane chains have stronger covalent bonds between molecules.
- (D) Shorter alkane chains are more polar and therefore boil at higher temperatures.

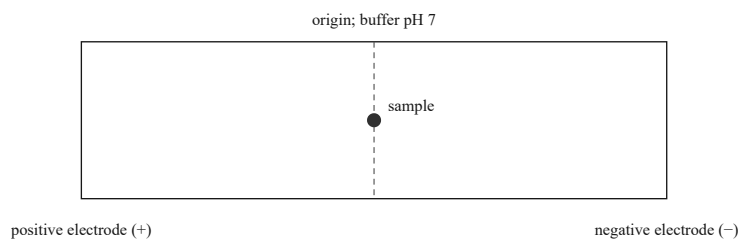
QUESTION 15

For $\text{Ag}_2\text{CO}_3(\text{s}) \rightleftharpoons 2\text{Ag}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$, $K_{sp} = 8.0 \times 10^{-12}$. Determine the molar solubility of Ag_2CO_3 in pure water. Assume hydrolysis of carbonate ions is negligible.

- (A) $2.0 \times 10^{-12} \text{ M}$
- (B) $2.8 \times 10^{-6} \text{ M}$
- (C) $1.3 \times 10^{-4} \text{ M}$
- (D) $2.0 \times 10^{-4} \text{ M}$

QUESTION 16

An amino acid has an isoelectric point of 2.8. It is placed in an electrophoresis buffer of pH 7.0. Which prediction is correct?



Electrophoresis field

- (A) It is positively charged and moves to the negative electrode.
- (B) It is neutral and remains at the origin because every amino acid is a zwitterion at pH 7.
- (C) It is negatively charged and moves to the negative electrode.
- (D) It is negatively charged and moves to the positive electrode.

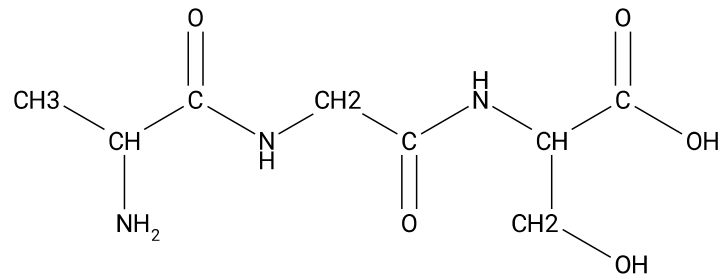
QUESTION 17

A buffer contains CH_3COOH and CH_3COO^- . A small amount of NaOH is added. Which statement best explains the response?

- (A) CH_3COOH reacts with added OH^- , so the pH increases only slightly.
- (B) CH_3COO^- releases OH^- , so the pH remains exactly constant.
- (C) The buffer stops ionisation completely, so no concentrations change.
- (D) Na^+ reacts with H^+ to form sodium metal.

QUESTION 18

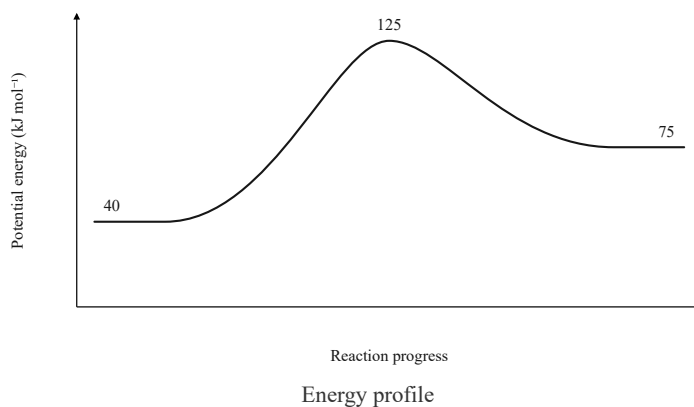
Determine the three-letter amino-acid sequence, read from the N-terminus to the C-terminus, for the displayed tripeptide.



- (A) Ser-Gly-Ala
- (B) Gly-Ala-Ser
- (C) Ala-Gly-Ser
- (D) Ala-Ser-Gly

QUESTION 19

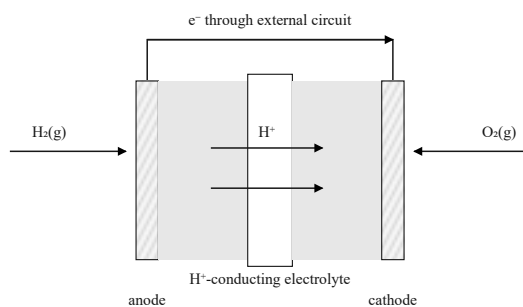
Use the energy profile to determine the activation energy of the reverse reaction.



- (A) 35 kJ mol⁻¹
- (B) 40 kJ mol⁻¹
- (C) 85 kJ mol⁻¹
- (D) 50 kJ mol⁻¹

QUESTION 20

Determine the cathode half-equation for a hydrogen fuel cell operating under acidic conditions.



Acidic hydrogen fuel cell

- (A) $2\text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^-$
- (B) $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\text{l})$
- (C) $2\text{H}_2(\text{g}) + 4\text{OH}^-(\text{aq}) \rightarrow 4\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$
- (D) $4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2(\text{g})$

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